

Absorption Modulation, Enhancement, and Narrowing Using Sub-Wavelength Gratings

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Abstract—The Sub-Wavelength Grating (SWG) has great potential to be used as a substrate in narrowband photodetectors of an adjustable detection range covered visible and near-infrared spectrum. By the combination of polycrystalline MAPbI₃ perovskite material and a specially designed SWG substrate, the absorption spectrum of perovskite material at specific wavelengths within inherently weak band-edge absorption would be enhanced due to guided-mode resonance effect, resulting in a modulated, enhanced, and narrowed absorption of perovskite material. We designed different parameters of SWGs to evaluate the absorption of perovskite within the bandgap of perovskite. The influences of depth, periodicity and fill factor are systematically studied using finite-difference time-domain (FDTD) simulation method. Finally, we achieved a double enhancement of absorption at 780 nm for SWG-perovskite sample compared with planar counterpart at which wavelength normally negligible absorption is witnessed. Moreover, we fabricated the designed SWG substrate and successfully deposited and infiltrated polycrystalline MAPbI₃ perovskite material on it.

Keywords—narrowband, sub-wavelength grating, FDTD, polycrystalline perovskite, photodetectors

I. INTRODUCTION

Perovskite materials have excellent optoelectrical properties such as long carrier diffusion length, high carrier mobility, high optoelectrical conversion efficiency, adjustable band gap and low fabrication cost. All these advantages make perovskite popular in photodetectors, solar cells and light-emitting diodes [1-5]. Based on the detection range, the photodetectors constructed by perovskite material could be divided into wideband photodetectors and narrowband photodetectors. Wideband photodetectors aiming at a long detectable wavelength range. Researches have been reported to enhance the efficiency [6, 7] and response speed [8, 9] of this

type of photodetectors. On the other hand, narrowband photodetectors tend to have a narrow spectral response with Full Width at Half Maximum (FWHM) less than 30 nm. This feature enables narrowband detectors to have important applications in biomedical sensing, hyperspectral imaging, military defense, and machine vision [10-12]. There are several methods for the preparation of narrowband perovskite photodetectors. Combining the optical filters with wideband perovskite photodetectors is one of the most widely adopted solutions [13-15]. However, this method requires extra bonding and packaging processes, which will increase the cost of fabrication. Adoption of narrowband optical absorption materials such as two-dimensional (2D) perovskite is another solution to narrowband photodetectors [16]. However, due to the large exciton binding energy of 2D perovskite materials, it is hard for carriers to be extracted, so the efficiency of the narrowband response is relatively low. For example, the detectivity of a photoconductor based on 2D (iso-BA)₂PbI₄ can only reach 4×10^{10} J under a high bias of 6 V [16]. Charge collection narrowing (CCN) is another common efficient solution for narrowband photodetectors [17-21]. The narrowing effect was firstly observed by Harrison *et al.* in an organic photovoltaic device in 1997 [22]. Then, in 2015, Paul's group has brought up the concept of CCN, realizing the first sub-100 nm FWHM, visible-blind red and NIR photodetectors [17]. By adjusting the thickness of the material and the bias voltage applied on the device, only the carriers generated by the deeply penetrated photons could be collected by the electrodes [1]. Nevertheless, the EQE of photodetectors based on CCN was only about 3%-23% [17-21]. By reducing the thickness of the internal perovskite active layer through an external perovskite layer to absorb incident photons of short wavelengths without generating photocurrent, the transmission loss decreases and the EQE of the

photodetector based on CCN can be improved to 23% [19]. Also, it will lose the ability of narrowband detection at a high bias voltage [17]. Thus, narrowband photodetectors with low cost of fabrication, high response efficiency and compatibility with high applied voltages are highly demanded. To reach the high response of narrowband detection regardless of high bias voltages, it is essential to realize narrowband absorption of the active material.

To achieve this, technologies including 2D photonics crystal structure [23], surface plasmon resonance [24] and optical cavity [25] have been applied. However, they have disadvantages such as high fabrication cost, complicated structure and relatively low efficiency. Another strategy for narrowband photodetectors is to construct polycrystalline perovskite combined with a sub-wavelength grating (SWG) substrate. The optical field would be modulated by the SWG structure, resulting in enhancing the absorption and suppressing the transmission and reflection of a specific wavelength of light within the inherently weak band-edge absorption spectrum without the requirement of thick perovskite material. Besides, by adjusting the composition of polycrystalline perovskite and different aimed detection wavelength at the absorption band-edge of perovskite, narrowband detection covering visible and near-infrared can be realized by changing the parameters of SWG.

In this work, we constructed planar devices and SWG-perovskite devices to compare the optical field distribution in these devices using finite-difference time-domain (FDTD) method to evaluate the absorption enhancement of SWG-perovskite device. A strongly enhanced optical field distribution in SWG-perovskite device confirmed the functionality of SWG. The influence of the depth, the periodicity and the fill factor of SWG on the reflectance/transmittance/absorption (R/T/A) of the perovskite layer were studied. Besides, the absorption spectra between the planar device and SWG-perovskite device are also compared. A nearly doubled absorption efficiency at the absorption band edge of the MAPbI₃ polycrystalline perovskite thin film is observed in the SWG-perovskite device compared with the planar device. We have also successfully deposited and infiltrated polycrystalline perovskite material into the SWG substrate. This work provided theoretical and experimental basis of SWG based narrowband photodetectors, which could be a promising solution to low-cost, high-quality narrowband photodetection.

II. WORKING PRINCIPLE AND STRUCTURE OF DEVICE

A. Working Principle of SWG

The basic theory of how SWG works in our devices to realize enhanced the narrowband absorption can be explained as followed. Any periodic variation in the refractive index would serve as a diffraction grating. SWGs are special type of diffraction gratings which the physical dimensions are less than the wavelength of incident light [26]. There is an important characteristic called guided-mode resonance (GMR) of SWG waveguide. GMR refers to rapid variations in the intensity of propagating waves that are diffracted by a waveguide grating. For example, by using SWG, a very high reflectivity ($R > 99\%$) at a specific wavelength based on GMR

effects in a poly-Si subwavelength periodic membrane is obtained [27]. Physically, GMR effects are due to coupling of the externally propagating diffracted fields to the guided modes of the waveguide causing resonance phenomenon, so that at the resonance frequency, a significant field localization effect occurs in the structure [28-30]. Different guided modes are coupled at a specific frequency, which can cause energy leakage in the structure and an efficient energy exchange between the reflected and transmitted waves occurs in small parameter ranges at resonance, which is beneficial to enhance the absorption of perovskite materials near the structure [27]. Through periodic modulations of the refractive index, the GMR effects can be realized in subwavelength waveguide grating structures. With the high reflectivity and narrowband nature of guided mode resonance effects, high reflectors, high transmitters and narrowband filters can be designed and manufactured. For SWGs, by rationally designing structural parameters and selecting appropriate incident conditions, the diffraction of the grating will produce narrowband, high diffraction efficiency, strong wavelength dependence and other properties. Thus, different parameters of SWG structures would cause the shift of the center wavelength of the reflection spectrum and the transmission spectrum [31]. Therefore, the center wavelength of narrowband absorption spectrum of the SWG-perovskite device can be changed.

B. Design of Planar and SWG-Perovskite Devices

The planar and SWG-perovskite devices are all based on the photoconductor structure as shown in Fig. 1. In conventional planar photodetectors, a large amount of light would be reflected by or transmitted through the active material, there is only part of the light is absorbed by the perovskite absorption layer as is shown in Fig. 1 (a). However, with SWG-perovskite structure working in the device, the reflected and transmitted light can be modulated by the SWG-perovskite structure as the substrate and a large amount of the reflected and transmitted light would stay in the perovskite absorption layer, resulting in an enhanced optical field, as is shown in Fig. 1 (b). Then this part of light would participate in the generation of photocarriers in the perovskite absorption layer, and enhance the response of the photodetectors. Moreover, the SWG-perovskite structure could only work with the designed size of the SWG aiming at a specific wavelength. Thus, when an active material with a wide band absorption spectrum is combined with a specially designed SWG, the absorption peak would be selected and narrowed.

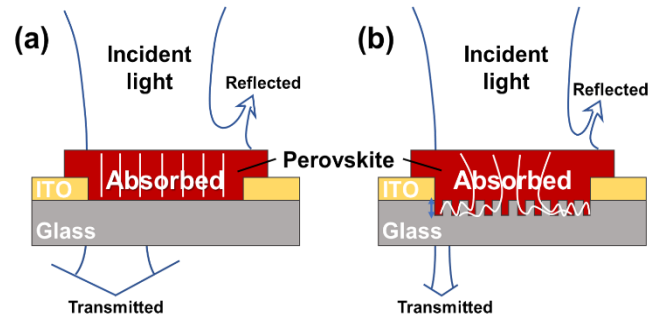


Fig. 1. Device structures and the working mechanism of the designed (a) planar device and (b) SWG-perovskite device.

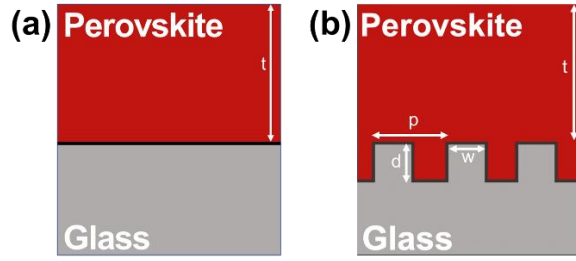


Fig. 2. Parameters of (a) planar device and (b) SWG-perovskite device

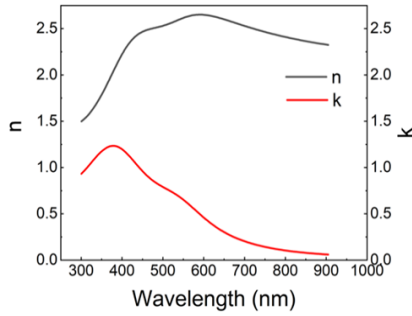


Fig. 3. Refractive index and extinction coefficient of MAPbI₃ polycrystalline perovskite film

The parameters of these two devices are shown in Fig. 2. The thickness of the perovskite absorption layer (t) is the same for both devices. In the SWG-perovskite device, there are additional parameters of the SWG structure including periodicity (p), depth (d) and fill factor (w/p) in Fig. 2 (b). Besides, the gap between the two electrodes of two photoconductors in Fig. 1 is both set as $5\ \mu\text{m}$, considering the manufacturing process and the working area. The perovskite absorption layer is set as $400\ \text{nm}$ for both devices. For the SWG-perovskite device, the perovskite material would fill the gaps of SWG first, then another $400\ \text{nm}$ of the polycrystalline perovskite thin film would be still deposited on the substrate.

III. SIMULATION OF OPTICAL FIELD DISTRIBUTION

Before the calculation and simulation works were processed, the basic parameters including the refractive index, extinction coefficient of the polycrystalline MAPbI₃ thin film used in the devices were measured by an ellipsometry. The refractive index and extinction coefficient of the polycrystalline MAPbI₃ film are shown in Fig. 3, which would be used for the simulation and verification works.

As is mentioned above, the SWG substrate could modulate, enhance and narrow the absorption of the material through the GMR effect by suppressing the reflected and transmitted light. Thus, the target wavelength of optimization should be located at a wavelength where the majority of light is reflected or transmitted and minority of light is absorbed. So, the target of the center wavelength of the narrowband detection was set as $780\ \text{nm}$, where the absorption of the polycrystalline MAPbI₃ film is weak and remains room for optimization. Firstly, all of the parameters of SWG (depth, periodicity and fill factor) were selected with resolution of $50\ \text{nm}$ for depth (from 50 to $300\ \text{nm}$), $50\ \text{nm}$ for periodicity (from 50 to $800\ \text{nm}$) and 0.1 for fill factor (from 0.1 to 0.9). By comparing the light absorption at

$780\ \text{nm}$ of these SWG-perovskite models, the SWG with depth of $300\ \text{nm}$, periodicity of $500\ \text{nm}$ and fill factor of 0.4 were selected for further optimization because they had the strongest absorption at $780\ \text{nm}$ compared with the other parameters. Then, these parameters were detailedly optimized with resolution of $10\ \text{nm}$ for depth (from 280 to $320\ \text{nm}$), $10\ \text{nm}$ for periodicity (from 480 to $520\ \text{nm}$) and 0.01 for fill factor (from 0.38 to 0.42). In the end, the best optimized parameter of the SWG-perovskite structure with the strongest enhanced light absorption at $780\ \text{nm}$ among all these structures with different parameters is listed in Table I.

TABLE I. OPTIMIZED PARAMETER OF SWG

	Depth (d)	Periodicity (p)	Fill factor (w/p)
Scale (nm)	290	510	0.39

To evaluate the performance of the designed SWG-perovskite structure, we carried out FDTD simulation to evaluate the light absorption of the perovskite absorption layer. Firstly, we input the refractive index and the extinction coefficient of the polycrystalline MAPbI₃ thin film and SiO₂, then modelled the planar and SWG-perovskite devices based on the structure shown in Fig. 3. We set the thickness of the polycrystalline MAPbI₃ thin film absorption layer to $t = 400\ \text{nm}$ for both structures, the substrate as SiO₂ for the planar device and the substrate of the SWG-perovskite structure in the SWG-perovskite device as SiO₂ and MAPbI₃ perovskite material filling inside the SWG. The parameters of the SWG-perovskite structure were set as $p = 510\ \text{nm}$, $d = 280\ \text{nm}$ and $w = 198\ \text{nm}$ as shown in Table I. Then, the optical field distributions of $780\ \text{nm}$ light for the planar and SWG-perovskite device are output by FDTD as shown in Fig. 4. It can be seen that the optical field intensity in the perovskite material of the planar device is much weaker than that of perovskite material in the SWG-perovskite device. Especially, for the perovskite material inside of the SWG, the optical field is strongly enhanced. Compared with the planar device, the light intensity of the perovskite absorption layer is redistributed in the SWG-perovskite device, causing a peak absorption at the band-edge of the perovskite. The optical field of $780\ \text{nm}$ is strongly enhanced because of the modulation of the GMR effect of the specially designed SWG [28]. The resonated light inside the SWG-perovskite device would cause an extended light path and finally resulted in localized enhancement of optical field, compared with the planar device. According to Lambert-Beer law as is shown in equation (1):

$$A = K \cdot l \cdot c \quad (1)$$

here, A is the absorbance of light, and K is the absorption coefficient, l is the distance of light passing through the material, c is the concentration of the material (g/L or mol/L). When other conditions are the same, the absorbance of light is proportional to the distance of light passing through the material. Therefore, it can be inferred that the longer light path in the SWG-perovskite device produced by the GMR effect would result in an enhanced absorption of the perovskite layer at $780\ \text{nm}$ even the thickness of the perovskite layer is only $400\ \text{nm}$.

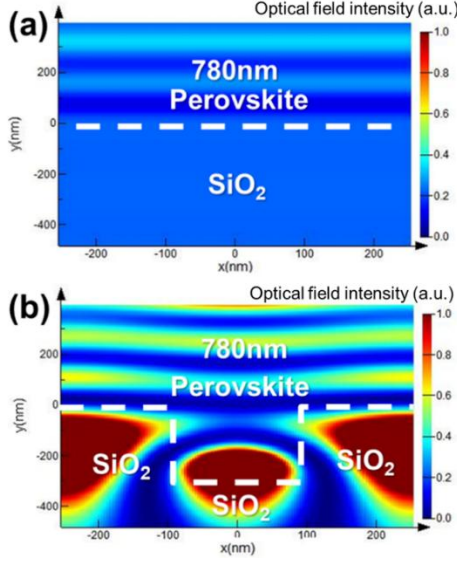


Fig. 4. Optical field distribution of (a) planar device and (b) SWG-perovskite device

To verify the absorption narrowing performance of the SWG-perovskite structure, the transmittance (T) and reflectance (R) are recorded in the simulation and the absorption (A) is calculated by $A = 1 - T - R$ [32]. The normalized power of T , R , A spectra of planar device and SWG-perovskite devices with different parameters represent their ability of light modulation, enhancement and narrowing, as is shown in Fig. 5. There is no narrowband absorption and the absorption of the planar perovskite follows the trend of absorption spectrum of the polycrystalline MAPbI₃ thin film. Light beyond 700 nm tends to be transmitted and reflected. Only 41% of the incident light is absorbed by the perovskite material at the wavelength of 780 nm. For the perovskite material with the enhancement of the designed SWG-perovskite structure, there is an abrupt rise in the absorption spectrum at aimed detection wavelength 780 nm with a FWHM of about 20 nm and the normalized power of light absorption is strongly enhanced from 0.41 to 0.95 at 780 nm. And the enhanced narrowband absorption of the device is realized by manipulating the transmission and the reflection of the 780 nm light. This enhancement of light absorption at 780 nm confirmed our assumption of absorption narrowing.

Moreover, we also investigated how the parameters affect the $R/T/A$ spectra of the SWG-perovskite device by sweeping the depth, periodicity and fill factor below and beyond the best optimized parameter. The results are plotted in Fig. 5(b-d). Changing the depth of the SWG-perovskite structure from 280 nm to 320 nm would slightly affect the red-shift of absorption peak from 770 nm to 790 nm. But the maximum of absorption peak of these SWG-perovskite structures with different depth only slightly varies from 0.94 to 0.95. And the FWHM of absorption spectra of these structures remain unchanged. The periodicity of the structure has the strongest effect on the absorption. A variety of periodicity from 480 nm to 520 nm could result in a large red-shift of absorption peak from 750 nm to over 800 nm. The intensity and FWHM of absorption would also be strongly suppressed to 0.85, when the periodicity is set

as 480 nm. Among all these parameters, the fill factor matters the least. Changing the fill factor from 0.38 to 0.42, the absorption peaks only red-shifted for several nanometers, the FWHM also remain unchanged. This provides important guidance during the fabrication of the SWG substrate, the periodicity should be carefully controlled, the depth and the fill factor have better robustness.

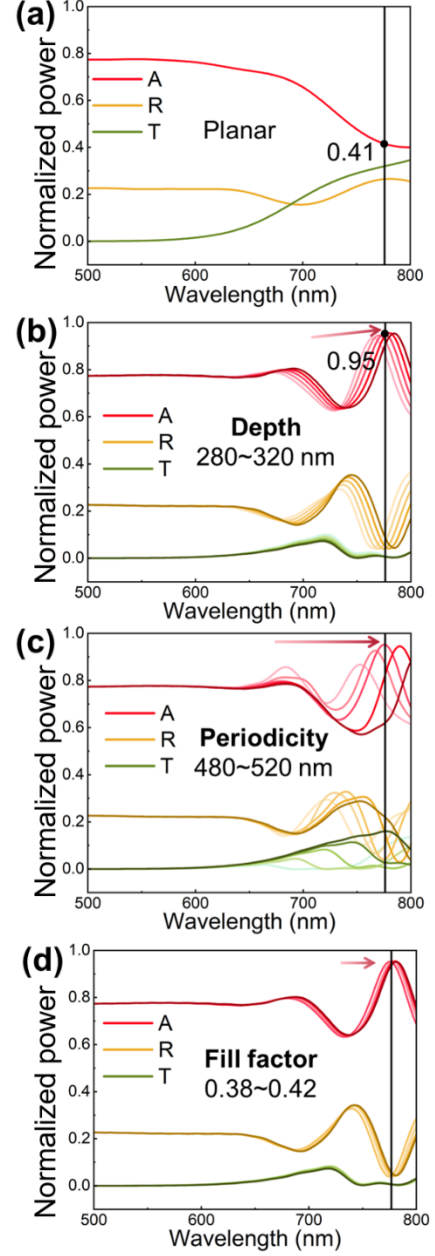


Fig. 5. R/T/A analysis of (a) planar device and SWG-perovskite devices with (b) different depth vary from 280 ~ 320 nm, the step is 10 nm; (c) different periodicity vary from 480 ~ 520 nm, the step is 10 nm; (d) different fill factor vary from 0.38 to 0.42, the step is 0.01.

IV. FABRICATION OF PEROVSKITE IN SWG

We used a Helium ion microscope focused ion beam (HIM-FIB) system from Zeiss to fabricate the designed SWG

structure. First of all, the ITO on the glass substrate was roughly removed and patterned by laser marking machine. Then, we used the Gallium ion FIB system to remove the ITO layer and create a $5\ \mu\text{m} \times 50\ \mu\text{m}$ area as the working area of photoconductors. After that, the Helium ion microscope was used to fabricate the SWG inside the working area. The ion beam energy, dwell time and pattern size were optimized to obtain the designed depth, period and fill factor of the designed SWG. Firstly, Ga ions with high beam current (2 nA) was used to clean the excess ITO remaining on the glass substrate during the laser printing process. Then, the ITO patterns connected with a narrow bar was etched to two separate electrodes with separation of $5\ \mu\text{m}$ and length of $50\ \mu\text{m}$. For now, ITO electrodes for the controlled group are fabricated. After this, a lower beam current (700 pA) of Ga ions was applied for SWG fabrications. The length and period could be directly set according to the simulated structures. However, the depth and width should be optimized with the dwell time and area. Because there could be a mismatch for the assigned width and fabricated width caused by the beam width and etching time of the Ga ions. The parameters including period, width and depth of SWGs were double checked using top and cross-sectional HIM before the final fabrication of SWGs. And the fabricated SWG structure is shown in Fig. 6 (a). After the fabrication, the SWG substrate were cleaned using ultrasound cleaner with alcohol.

The fabricated SWG substrate is shown in Fig. 6 (a). This figure is obtained from a tilted sample stage; thus, the vertical and horizontal scale are different. Thanks to the high manufacturing precision up to 5 nm of HIM-FIB, the SWG substrate is successfully fabricated. As we can see, the mismatch between the designed parameters with the fabricated parameters of the SWG structure is less than ten nanometers.

Then, the polycrystalline MAPbI₃ thin film was spin-coated onto the SWG structure and annealed for 10 minutes. The cross section of the device was characterized using HIM-FIB as shown in Fig. 6 (b), which indicates that the polycrystalline MAPbI₃ perovskite has successfully filled the nanometer-size gaps of the SWG substrate.

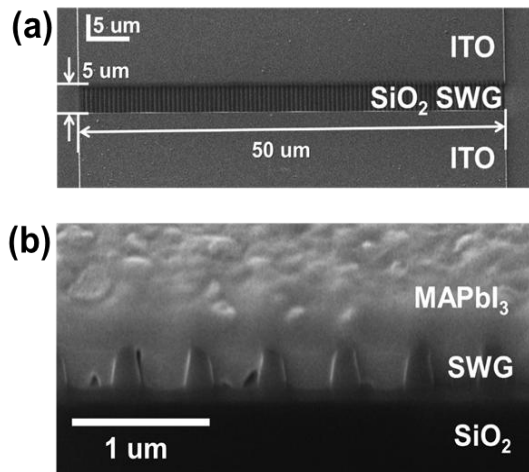


Fig. 6. Tilted-stage SEM image of (a) fabricated SWG substrate and (b) polycrystalline MAPbI₃ perovskite material deposited in SWG substrate.

V. CONCLUSION

Based on the redistribution and narrowing effect of SWG-perovskite structure constructed by SiO₂ and the MAPbI₃ perovskite material, we have evaluated the theoretical possibility of the realization of narrowband photodetectors using FDTD simulation. The working principle of the SWG is illustrated and the optimization process of depth, periodicity and fill factor of the SWG-perovskite structure are evaluated. Aiming at narrowband detection wavelength of 780 nm, the parameters of SWG-perovskite structure are optimized and the best performed one have the depth of 290 nm, periodicity of 510 nm and fill factor of 0.39. We compared the difference of the optical field distribution between the planar device and the SWG-perovskite device. The strongly enhanced optical field in the SWG-perovskite device indicates the successful generation of the GMR effect. Finally, the absorption of the SWG-perovskite device at the weak absorption band-edge, 780 nm, is 0.95, which is over doubled than that of the planar device (0.41). Also, the FWHM of it is narrowed to 20 nm. The comparison of the optical field intensity in the planar perovskite material and the perovskite material of the SWG-perovskite device has been made to prove the optical field intensity at 780 nm in the perovskite material of SWG-perovskite device is strongly enhanced. Besides, the impact of depth, periodicity and fill factor of the SWG-perovskite structure on the R/T/A of the perovskite layer is evaluated and studied. Beyond this, we fabricated the designed a SWG substrate using a HIM-FIB system. This process could be simplified by nano-imprinting technology in the future to minimize the cost of fabrication. The polycrystalline perovskite material MAPbI₃ has been successfully deposited into the SWG substrate. This method provides a candidate solution to low cost, high performance and adjustable narrowband photodetectors based on polycrystalline perovskite materials.

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